

Breaking Conjugacy: Logistic Regression & Newton's Method

INFO 521 · student notes · Module 5 · Lecture A · ~7 min read

Module 4 ended with everything Gaussian and every formula closed-form. Asking the clinical question — *is this patient hypertensive?* — binarizes the outcome and breaks that machinery. This is the second act of the course, and where Part 2 opens.

The conjugacy break

The target is now a **Bernoulli** outcome, not a Gaussian one. A Gaussian prior on $\mathbf{w}$ meeting a Bernoulli likelihood has **no conjugate pair** — the closed-form posterior of Module 4 is gone. Same data ($\approx 39\%$ prevalence), new target type, and the algebra that made Bayes easy is structurally unavailable.

Logistic regression

Keep the linear score and squash it into a probability with the **sigmoid**,

$$p(y = 1 \mid \mathbf{x}, \mathbf{w}) = \sigma(\mathbf{w}^\top \mathbf{x}), \quad \sigma(a) = \frac{1}{1 + e^{-a}}.$$

It is still linear in $\mathbf{w}$ (basis functions still apply) and **discriminative** (it models $p(y \mid \mathbf{x})$ directly); the decision boundary $\mathbf{w}^\top \mathbf{x} = 0$ is still a line. The log-likelihood is the cross-entropy, and setting its gradient to zero gives $\mathbf{X}^\top (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$ — which *looks* like the normal equations, but $\boldsymbol{\mu}$ hides $\mathbf{w}$ inside the sigmoid. There is **no closed form**: the equation is transcendental, not merely harder. So we iterate.

Newton–Raphson

Newton repeatedly jumps to the peak of the local quadratic, using both the gradient (which way is up) and the **Hessian** (how the slope curves): $\mathbf{H} = -\mathbf{X}^\top \mathbf{R} \mathbf{X}$ with $\mathbf{R} = \text{diag}(\mu_n(1 - \mu_n))$. It is negative definite, so the log-likelihood is concave and the peak Newton finds is *the* peak — Module 3's uniqueness argument, with $\mathbf{X}^\top \mathbf{X}$ reweighted into $\mathbf{X}^\top \mathbf{R} \mathbf{X}$. The fence-sitters near $\mu = \frac{1}{2}$ carry the most weight — they locate the boundary. On all of NHANES it converges in about five steps to $\hat{\mathbf{w}} \approx (-0.49, 0.52)$: each SD of age multiplies the odds of hypertension by $e^{0.52} \approx 1.7$, and the fitted probability crosses $\frac{1}{2}$ near age 66.

The logistic explorer (framed as tumor size, benign vs malignant — the same model) lets you set the bias and steepness by eye and watch the log loss and accuracy respond, then **Show MLE fit** snaps to the maximum-likelihood solution and **Generative comparison** shows a per-class Gaussian model reaching essentially the same boundary.

What survived, and what did not

The frequentist story is intact: $\hat{\mathbf{w}}$ by Newton, uniqueness from concavity, even Module-3-style uncertainty from the curvature at the peak. The Bayesian story took the hit: the posterior is known

only up to its normalizer (the evidence integral has no closed form), so exact credible intervals and honest predictive probabilities are out of reach. **Optimization survived binarization; integration did not.**

What to remember

- Binarizing gives a Bernoulli likelihood; with a Gaussian prior there is no conjugate posterior.
- Logistic regression: $p(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x})$, concave ℓ , gradient $\mathbf{X}^\top (\mathbf{y} - \boldsymbol{\mu})$.
- Newton uses curvature ($\mathbf{H} = -\mathbf{X}^\top \mathbf{R} \mathbf{X}$) and converges in ~ 5 steps.
- Finding the peak is easy; knowing the mass around it is the hard part (Lecture B).

Through-lines

$\mathbf{X}^\top \mathbf{X}$ becomes $\mathbf{X}^\top \mathbf{R} \mathbf{X}$, and the Hessian at the peak is about to become the covariance of a Gaussian approximation to the posterior — Lecture B reuses it directly.